

Global Internet Adoption Analysis (1990–2025): A Comparative Study of Selected Countries Using World Bank World Development Indicators

1. Project Objective

This project aims to analyze internet adoption trends across selected countries between 1990 and 2025 using the World Bank World Development Indicators (WDI) dataset. The objective is to identify adoption patterns, compare countries, evaluate long-term growth, and communicate findings through an interactive Power BI dashboard that supports data-driven decision-making.

2. Problem Statement

Internet access has become one of the strongest indicators of technological development and digital inclusion. While developed countries have achieved high internet penetration, many developing countries continue to experience slower adoption rates due to economic, infrastructural, and policy-related challenges.

This project investigates how internet adoption has evolved over time across ten selected countries and highlights the differences in digital growth using publicly available World Bank data.

3. Dataset Description

Dataset: World Development Indicators (WDI)

Source: World Bank

File Used: WDICSV.csv

The World Development Indicators dataset contains hundreds of development indicators covering more than 200 countries and territories across several sectors including:

- Economy
- Health
- Education

- Environment
- Infrastructure
- Technology
- Trade
- Governance

For this project, the analysis focused on the technology sector using the indicator; Individuals using the Internet (% of population)

Ten representative countries were selected:

- Brazil
- Canada
- China
- Germany
- India
- Japan
- Nigeria
- South Africa
- United Kingdom
- United States

4. Data Cleaning Process

The original dataset contained 396,970 rows and 70 columns, with yearly values spanning 1960 to 2025.

The following data preparation steps were performed using Python and Pandas:

Step 1;

Loaded the World Development Indicators dataset into Python using Pandas.

Step 2;

Inspected the dataset by:

- Viewing the first five rows
- Checking the dataset dimensions
- Examining column names
- Reviewing data types
- Identifying missing values

Step 3;

Filtered the dataset to retain only indicators related to internet adoption by searching for Internet-related indicators.

Step 4;

Selected the primary indicator; Individuals using the Internet (% of population), to simplify the analysis and maintain consistency.

Step 5;

Filtered the dataset to include only ten representative countries:

- Brazil
- Canada
- China
- Germany
- India
- Japan
- Nigeria
- South Africa
- United Kingdom
- United States

Step 6;

Excluded regional aggregates and economic groups such as:

- World
- High income
- Low income
- OECD
- European Union

to ensure the analysis focused only on individual countries.

Step 7;

Reshaped the dataset from wide format into long format using the “melt()” function, creating two new columns:

- Year
- Value

Step 8;

Converted the Year column into an integer data type.

Step 9;

Ensured the Value column remained numeric for statistical analysis.

Step 10;

Removed rows containing missing values to improve data quality and ensure accurate visualization.

After cleaning, the dataset contained 674 valid observations across the selected countries and years.

Step 11;

Exported the cleaned dataset as “Cleaned_Internet_Indicators.csv” for visualization in Microsoft Power BI.

5. Analysis Methodology

The project followed a structured data analytics workflow involving data collection, cleaning, transformation, analysis, visualization, and interpretation.

The World Development Indicators (WDI) dataset was imported into Python using the Pandas library for data preparation. Initial exploration included examining the dataset structure, dimensions, column names, data types, and missing values.

To align the analysis with the project objective, internet-related indicators were identified and filtered. The analysis focused on the indicator "Individuals using the Internet (% of population)", as it provides a consistent measure of internet adoption across countries.

Ten representative countries (Brazil, Canada, China, Germany, India, Japan, Nigeria, South Africa, the United Kingdom, and the United States) were selected for comparative analysis, while regional aggregates and economic groupings were excluded.

The dataset was first transformed from wide format to long format using the "melt()" function, making it suitable for time-series analysis and visualization. Missing values were removed, appropriate data types were assigned, and the cleaned dataset was exported for dashboard development.

Microsoft Power BI was used to create an interactive dashboard featuring KPI cards, charts, and slicers to facilitate comparative analysis and trend visualization.

6. Dashboard Overview

An interactive dashboard was developed in Microsoft Power BI to present internet adoption trends across the selected countries from 1990 to 2025.

The dashboard includes:

- KPI Cards displaying:
 - Highest Internet Adoption
 - Lowest Internet Adoption (Non-Zero)
 - Average Internet Adoption
 - Number of Countries Analysed
- Interactive slicers for:
 - Year

- Country
- Visualizations including:
 - Area Chart showing average internet adoption by country
 - Pie Chart illustrating each country's share of total internet adoption
 - Clustered Bar Chart displaying cumulative internet adoption by country
 - Line Chart illustrating internet adoption trends over time

These interactive elements allow users to explore internet adoption patterns by country and across different years.

7. Key Findings

The analysis revealed several important findings regarding internet adoption among the selected countries:

- Canada recorded the highest average internet adoption rate among the selected countries over the study period.
- Nigeria recorded the lowest average internet adoption rate, indicating comparatively slower digital growth.
- Internet adoption remained relatively low during the 1990s and early 2000s but increased rapidly from the mid-2010s onward.
- Developed countries, including Canada, Germany, the United Kingdom, and the United States, consistently maintained higher internet adoption rates than developing countries.
- Although developing countries such as Brazil, China, India, and South Africa experienced substantial growth, their adoption rates remained below those of the leading developed economies throughout most of the study period.

8. Insights

The analysis demonstrates the significant progress made globally in internet adoption over the past three decades while highlighting persistent disparities between developed and developing economies.

Countries with stronger economic development and digital infrastructure generally achieved earlier and higher levels of internet adoption. Conversely, developing countries experienced slower initial growth but showed remarkable improvements in more recent years, reflecting increasing investments in digital infrastructure and expanding internet accessibility.

The findings also indicate that internet adoption has accelerated considerably in the last decade, emphasizing the growing importance of digital technologies in economic development, education, communication, healthcare, and business operations.

9. Recommendations

Based on the findings of this study, the following recommendations are proposed:

- Governments should continue investing in digital infrastructure to improve internet accessibility, particularly in underserved and rural communities.
- Policies aimed at reducing the cost of internet services should be strengthened to increase affordability and encourage wider adoption.
- Investments in digital literacy programmes should be expanded to equip citizens with the skills required to benefit from internet technologies.
- Public-private partnerships should be encouraged to accelerate broadband expansion and reduce the digital divide between developed and developing economies.
- Continuous monitoring of internet adoption indicators should be maintained to support evidence-based policy formulation and national digital transformation initiatives.

10. Conclusion

This project successfully applied the complete data analytics workflow to examine global internet adoption trends using the World Bank World Development Indicators dataset.

Through data cleaning, transformation, visualization, and analysis, the project identified clear differences in internet adoption across selected countries and demonstrated the rapid growth of internet usage over recent decades.

The interactive Power BI dashboard provides an effective tool for exploring internet adoption trends and comparing countries over time. The findings generated from this analysis can support policymakers, researchers, and development practitioners in understanding digital inclusion and identifying opportunities for improving internet accessibility.